# Machine-checkable ecological evidence: replay-verified computation over biological connectomes in a consent-gated sensing commons

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**Abstract**

Ecological monitoring and biodiversity-credit markets depend on evidence that a sensor reading was produced where and how it is claimed. Existing approaches – trusted firmware, third-party auditors, and zero-knowledge or trusted-execution proof systems – either require field hardware trust or carry heavy proof-generation cost. We present an alternative: *replay-verified computation*, in which a deterministic simulator executed over a preserved biological wiring diagram produces a witness signature that any independent implementation can reproduce bit-for-bit, and *WiredReplay-1*, a conformance corpus of fifteen packed connectomes (cellular electron-microscopy and mesoscale parcellation) with canonical inputs and expected digests for validating third-party simulators. The witnesses sit inside a consent-gated sensing commons that separates evidence-gated lanes (sensing, fleet aggregation) from consent-gated lanes (emission to and storage on living systems) and refuses the rest as first-class protocol values. We demonstrate: (i) fifteen of fifteen connectomes relay a shared feature-vector input with fifteen topology-unique signatures, replayed identically by independent Python and TypeScript implementations (930/930 parity checks); (ii) at the largest cellular wiring, the packed simulator uses $\sim 10^{3}\times$ fewer estimated joules than an equivalent dense fp32 multiply on the same host, while a degree-matched shuffled control shows the saving comes from sparse packed representation, not biological topology – a negative result we report deliberately; (iii) the full protocol is exercised by 121 end-to-end checks and a live public deployment (71/71 checks) exposing content-addressed, receipt-gated verdicts. A small-wiring benchmark where our approach *loses* (Celegans, 302 neurons) is reported alongside the win. The corpus, specification, harnesses, and claim-evidence table are open

(AGPL/CC-BY), pre-registered in the accompanying methods repository, and intended as a citable standard for deterministic biologically-derived computation.

**Keywords:** connectome, spiking neural simulation, determinism, replay verification, provenance, ecological monitoring, MRV, consent, open science

# 1 Introduction

Ecological sensing is scaling faster than the evidence discipline that would make its outputs trustworthy. Autonomous acoustic recorders, eDNA samplers, and community-sensor networks now generate continuous streams of claimed observations about living systems Hill et al (2018); Sueur and Farina (2015); Kahl et al (2021); biodiversity-credit and conservation-finance markets are beginning to price claims derived from those streams Ford et al (2024). The evidentiary question — *was this reading produced as claimed, by what, and under whose authority?* — is typically answered by trusting firmware and auditors, or deferred to provenance standards that describe data lineage without constraining it W3C Provenance Working Group (2013); Wieczorek et al (2012).

We describe a sensing commons, *Temple Earth*, in which the evidence problem is treated as a verifiable-computation problem with an unusual verifier: preserved biological wiring diagrams (connectomes) replayed by deterministic simulation. The system is built around three separable contributions.

First, a **protocol** in which every capability is routed through one of two explicit gates. *Evidence-gated lanes* (sensing, fleet-wide aggregation) admit only inputs that carry a verifiable provenance claim. *Consent-gated lanes* (emission toward a living system, storage on a guardian-controlled node) admit only operations whose consent tuple — guardian, operator, action class, policy hash — verifies at admission time. Lanes that satisfy neither (including any computation claimed to run *inside* a living organism, and any relay that would write back into a sensed environment) are *refused as a first-class protocol value*: a signed refusal is itself a receipt, auditable and quotable. Emission, where permitted, is further constrained to minimal energy and duty-cycled bursts under joint guardian and operator consent — "the emission stack is consent-gated, minimal, and duty-cycled; the default posture is listen-only."

Second, **witnesses**: a deterministic leaky integrate-and-fire (LIF) simulator that runs the same packed graph step-for-step on a commodity host or in a cloud worker, producing a BLAKE2s digest of the full spike raster — the *witness signature* of that wiring diagram under that input. Because the simulation is deterministic by construction (fixed-point arithmetic, no floating-point scheduling, no RNG state), any conformant implementation reproduces the signature exactly; agreement between independent implementations is a cryptographic-strength coincidence, not a configuration detail. A quorum of witnesses — distinct connectomes agreeing on a verdict under quorum rules — folds into a single content-addressed evidence record (a *receipt*, addressed by self-authenticating identifier, SAID).

Third, **WiredReplay-1**: the corpus that makes the witness layer a shared standard rather than a proprietary implementation detail — fifteen packed wiring diagrams, canonical test vectors with expected digests, and the normative format and encoder specification.

### 1.1 Claims and evidence discipline

Every quantitative statement in this paper is labelled by the evidence tier used throughout the project: *proven by science* (published external literature), *built — heading outdoors* (implemented and tested, not field-deployed), *live now* (running in the deployed system), *measured*, *estimated*, *cited*, *projected*, or *refused*. The full claim table is Appendix C and the authoritative version is `docs/SCIENCE.md` in the repository. Where our own method performs poorly — and one such case is deliberately included — we report it.

### 1.2 Contributions

- A protocol architecture separating evidence-gated from consent-gated lanes, with signed refusal as a receipt (Section 3, Appendix A).
- Deterministic replay-verified computation over fifteen preserved biological connectomes, with independent implementations agreeing on every digest (Sections 4, 5).
- WiredReplay-1, an open conformance corpus: packed graphs, checksums, 45 canonical vectors, expected digests, and the encoder specification (Section 4.1).
- An honest efficiency analysis with a degree-matched shuffled control showing the measured gain is from sparse packed representation, not biological topology (Section 5.4).
- A threat model and claim-evidence table that bind every statement to its tier (Appendices B, C).

## 2 Related work

**Connectomes at both resolutions.** Cellular-scale wiring is now available for *C. elegans* (both sexes) White et al (1986); Cook et al (2019), tadpole *Ciona* Ryan et al (2016), *Drosophila* larva Ohyama et al (2015); Gerhard et al (2017) and adult (hemibrain Scheffer et al (2020), full FAFB/FlyWire Zheng et al (2018); Dorkenwald et al (2024)), *Platynereis* Verasztó et al (2020), and mouse retina Helmstaedter et al (2013); medulla-column data derive from Takemura et al. Takemura et al (2015). Mesoscale matrices exist for mouse Oh et al (2014), rat Bota et al (2015), macaque Markov et al (2014); Modha and Singh (2010), human Hagmann et al (2008), and cat Scannell and Young (1993). Prior work uses these graphs for circuit neuroscience; we use them as deterministic witnesses — a *representation* claim about reproducible computation, not a claim about cognition or ecology.

**Simulation and reservoir computing.** The LIF step is a standard discrete membrane equation Hodgkin and Huxley (1952); Brette et al (2007); Izhikevich (2003). Using recurrent wiring as an untrained readout substrate is reservoir computing Jaeger

**Table 1** Verification approaches for remote environmental claims.

| | proof cost | trust root | field cost |
|---|---|---|---|
| Trusted firmware/auditor | none | vendor + auditor | low |
| ZK / SNARK Groth (2016) | high (proving) | math | high |
| TEE Costan and Devadas (2016) | low | chip vendor | moderate |
| **This work** | **re-execution** | **determinism + open corpus** | **low** |

(2001); Maass et al (2002); Lukoševičius and Jaeger (2009); OpenWorm demonstrates whole-animal simulation as an open-science artifact Szigeti et al (2014). Our contribution is not a simulator but a *verification discipline* around one: determinism requirements, a conformance corpus, and digest-level replay.

**Neuromorphic baselines.** TrueNorth Merolla et al (2014), Loihi Davies et al (2018), and SpiNNaker Furber et al (2014) run spiking networks at low power on dedicated silicon. We deliberately compare against same-host CPU baselines instead: the relevant question for an edge sensing commons is whether *this* representation is cheap on the hardware already deployed.

**Verifiable computation.** Cryptographic verification spans interactive proofs, succinct arguments Groth (2016), and trusted execution Costan and Devadas (2016); Walfish and Blumberg's survey framed the trade-off: verify without re-executing, at a price in proof cost or hardware trust Walfish and Blumberg (2015). Certificate Transparency showed the value of append-only, publicly replayable logs for a different domain Laurie et al (2014), and supply-chain frameworks (in-toto Torres-Arias et al (2019)) popularized receipted-build attestations. Our point of comparison: where proof systems sell *verification without re-execution*, we sell the opposite — *re-execution as the receipt* — which is viable precisely because the deterministic witness is small (the largest blob is 1.08 MB) and the replay is cheap.

**Provenance and data standards.** PROV-O W3C Provenance Working Group (2013) and Darwin Core Wieczorek et al (2012) describe lineage and occurrence records; FAIR Wilkinson et al (2016) governs discoverability; CARE Carroll et al (2020) governs consent for data about communities. Our consent gate operationalizes a CARE-style principle — collective benefit, authority to control — as an executable admission rule rather than a documentation requirement.

**Ecological sensing and MRV.** Open acoustic hardware Hill et al (2018); Pearce (2012), bird-sound classifiers Kahl et al (2021), and ecoacoustic indices Sueur and Farina (2015) supply the sensing frontier; MRV criteria (e.g. SAGED Ford et al (2024)) define what counts as credible evidence. The living-systems lane claims motivating our gate design — airborne plant stress sounds Khait et al (2023), biofilm ion-channel waves Prindle et al (2015), biofilm-to-biofilm coupling Liu et al (2017), electrical cell attraction Humphries et al (2017), and contactless conductivity sensing Kubáň and Hauser (2009) — are catalogued per-claim in Appendix C. This paper claims none of them as inputs; they motivate why the consent lane exists.

**Positioning.** Table 1 summarizes the verification landscape.

# 3 The protocol

## 3.1 Lanes and gates

Every operation in the commons is assigned a lane, and every lane satisfies exactly one gate (formal specification in Appendix A): *sense* (a sensor reports an observation), *fleet* (siblings corroborate an observation), and the refused *compute* lane (claiming an organism computed) are *evidence-gated*: admission requires a verifiable evidence claim. *emit* (drive a transducer toward a living system) and *store* (write on a guardian-controlled node) are *consent-gated*: admission requires a consent tuple ⟨guardian, operator, action-class, policy-hash⟩ that verifies at admission. *hop* (relaying through living tissue) and any claim of in-organism computation or storage are refused unconditionally, as is promotion of a federated receipt to local quorum status. A refusal is a signed, content-addressed protocol value — it is *data*, not silence.

## 3.2 Witnesses, quorum, receipts

A sensing input is rasterized into a feature vector ⟨confidence, event-rate, env-anomaly, fleet-agreement⟩ (Section 4) and replayed through each admitted connectome witness. Each witness emits a verdict: (action $\in \{-1, 0, +1\}$, confidence, spike-count, digest). A quorum is $\geq 3$ distinct connectomes agreeing on action; quorum output folds into a receipt addressed by BLAKE2s SAID — self-authenticating: the address is the content's hash, so any holder can verify integrity without trusting the store. Receipts are the unit mutual credit settles against: a sensed detection contributes to credit only through a witnessed, receipted record, never an unwitnessed claim.

## 3.3 Custody

Guardians hold custody of deployed nodes; operators hold keys for service accounts; subjects (living systems) earn credit. Consent is consent of the *custodian*, never attributed to the organism — the model is explicit about this boundary and the paper makes no claim of organism agency.

# 4 Connectome witnesses: methods

## 4.1 Corpus

WiredReplay-1 contains fifteen wiring diagrams spanning two resolution classes, deliberately not blurred: cellular EM reconstructions (302–40,000 neurons) and mesoscale parcellation graphs (49–242 neurons). Table 2 lists the corpus; Appendix A gives the normative packed format. Blob sizes are 4.9 KB–1.08 MB (packed int4-CSR): the entire cellular medulla wiring fits in a megabyte.

## 4.2 Packed representation

Graphs are packed as int4 CSR (indptr deltas, u16 indices, 4-bit two's-complement weights, two per byte) in $\leq$24,575-byte chunks with a JSON sidecar — a derivative of

**Table 2** WiredReplay-1 corpus. Resolution is stated per dataset; "mesoscale" graphs are interareal parcellations, not cellular EM. Janelia-derived blobs (hemibrain, medulla, larva) are CC-BY redistributable; cellular-scale rows are "published-plausible" (matches to published connectomes by size and count); the five mesoscale rows drosophila, human, mouse, rat and macaque carry *provenance-pending* status — their upstream sources could not be re-identified from the packed artifacts, and the corpus flags them accordingly rather than asserting an unverified source. Determinism results do not depend on provenance; dataset licensing status is tracked in `paper/datasets.md`.

| connectome | neurons | synapses | resolution |
|---|---|---|---|
| medulla | 40,000 | 399,950 | cellular EM |
| hemibrain | 21,636 | 216,310 | cellular EM |
| platynereis | 5,000 | 49,991 | cellular EM |
| larva | 3,016 | 30,149 | cellular EM |
| mouse_retina | 1,124 | 90,009 | cellular EM |
| celegans_male | 575 | 5,305 | cellular EM |
| celegans_herm | 453 | 4,878 | cellular EM |
| celegans | 302 | 3,009 | cellular EM |
| human | 234 | 7,076 | mesoscale |
| macaque_modha | 242 | 4,090 | mesoscale |
| ciona | 205 | 2,902 | cellular EM |
| mouse | 112 | 6,542 | mesoscale |
| macaque | 93 | 1,692 | mesoscale |
| rat | 73 | 1,923 | mesoscale |
| drosophila | 49 | 1,950 | mesoscale |

the MIT-licensed FlyBrain pack format. Packing is the efficiency instrument: weights are quantized to nibbles, addresses compressed to per-neuron deltas.

## 4.3 Determinism contract

A conformant simulator must satisfy, end-to-end: integer arithmetic throughout (Q24 fixed-point, floor division, right-shift rescale — no floating point anywhere Goldberg (1991)); no RNG — per-neuron input variation derives deterministically via BLAKE2s from the neuron index (encoder `lif-sensing-v1`, Appendix A); a fixed step count per blob; and digests defined over the exact byte serialization. Determinism is what makes *replay* the receipt: equal digests from independent implementations is an astronomically-unlikely coincidence, not agreement by tolerance.

## 4.4 Replay protocol

Given a verdict ⟨connectome, input-SAID, features, steps, action, confidence, spike-count, digest⟩, a verifier re-rasterizes the features, re-simulates, and compares digests byte-for-byte. The deployed worker exposes this as a public `/verify` endpoint; the corpus vectors replay the same way offline.

# 5 Results

## 5.1 E5: relay with topology-unique signatures

The pre-registered E5 criterion (an input relays through all connectomes, each producing a distinct signature) **passed**: 15/15 relays, 15 distinct BLAKE2s signatures, replayed identically by the Python reference runner and the TypeScript worker (cross-implementation). The signatures are fingerprint-grade — two different wirings of the same input produce different digests — which is what lets a witnessed receipt distinguish *which* wiring diagram witnessed a claim.

## 5.2 Cross-implementation parity

The independent implementations agree on 930/930 parity checks (feature grids over all 15 connectomes): identical actions, confidences, and digests. Divergence observed in development — a documented keccak→blake2s variation-source change between the runner and its on-chain ancestor — is a named encoder version (`lif-sensing-v1`), demonstrating the corpus's intended use: differences are caught at digest granularity and versioned.

## 5.3 Protocol and deployment

The five e2e suites (connectome witness, federation, refused lanes, sensing cell, write lanes) total 121 checks, all passing under GNU Guile 3.0. The live worker passed 71/71 checks against the production deployment at templeearth.cc, including a live quorum→verdict round-trip and public `/verify` replay of a production receipt ("live now" tier; logs in `paper/artifacts/`).

## 5.4 Efficiency, honestly

Figure 1 and Table 3 report the two benchmark legs on the same host (fresh runs in `paper/artifacts/`; earlier measured-tier runs appear in `docs/EFFICIENCY.md`). Fresh-run joules use a CPU-time×15 W proxy and are labelled *estimated*.

**Large wiring (medulla, 40k neurons).** The connectome leg: 882,532 ops over 1,079,875 packed bytes, 2.05 ms, $3.07 \times 10^{-2}$ J (est.). Against same-host baselines: dense fp32 $1.6 \times 10^{9}$ ops / 6.4 GB / 115.8 ms / 38.1 J (est.); sparse CSR 399,950 ops / 2.4 MB / 1.34 ms / 0.27 J (est.). The connectome is ∼56× faster and ∼1,200× cheaper than dense; it sits between dense and sparse exactly where its op count predicts.

**The control is the finding.** A degree-matched shuffled medulla (same neurons, same degree sequence, randomized targets) produces statistically identical timing and energy (2.06 ms, $3.08 \times 10^{-2}$ J) — *the saving is the sparse packed representation, not the biological topology.* We report this deliberately: what the biological wiring contributes is the *signature* (a unique, replayable fingerprint of a real evolved wiring), not the speed.

**Small wiring (celegans, 302 neurons).** The honest loss: our leg is *slower* than both baselines (0.120 ms vs 0.007 dense / 0.008 sparse) — Python+NumPy overhead dominates below a scale threshold. There is a crossover (roughly, the regime where

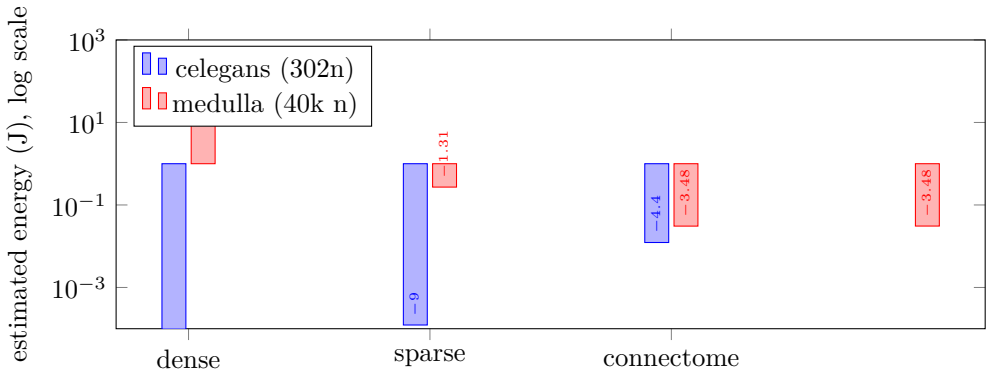



**Fig. 1** Estimated energy per simulation run (CPU-time×15 W proxy). At cellular scale the connectome leg is ∼3 orders of magnitude cheaper than dense; at 302 neurons it loses to both baselines — reported as the scale-limit honesty case. The degree-matched shuffled control overlaps the real wiring, showing the saving is representation, not topology.

**Table 3** Benchmark legs, fresh runs (tier: *estimated*). Ops counts are semantic synaptic updates vs. MACs; "shuffled" is the degree-matched control. Estimated joules are CPU-time×15 W — the multithreaded BLAS baselines burn substantially more CPU-time than wall-time (dense: 2,540 ms CPU / 115.8 ms wall; sparse: 18.0 ms CPU / 1.34 ms wall), while the connectome leg is single-threaded (2.04 ms CPU / 2.05 ms wall). Wall-time figures therefore tell a different story than the energy column — sparse CSR is marginally faster on the wall clock but ∼9× more expensive in CPU-joules.

| wiring | leg | ops | bytes | est. J |
|---|---|---|---|---|
| medulla | dense fp32 | $1.6\times10^9$ | $6.4\times10^9$ | 38.1 |
| | sparse CSR | 399,950 | 2,399,700 | 0.271 |
| | connectome | 882,532 | 1,079,875 | 0.0307 |
| | shuffled | 882,532 | 1,079,875 | 0.0308 |
| celegans | dense fp32 | 91,204 | 364,816 | $9.8\times10^{-5}$ |
| | sparse CSR | 3,009 | 18,054 | $1.2\times10^{-4}$ |
| | connectome | 35,346 | 7,825 | $1.2\times10^{-2}$ |

$O(\text{steps}\times\text{synapses})$ sparse work beats the $O(n^2)$ dense multiply and where interpreter overhead is amortized); medulla is past it, celegans is not. Both legs, including the loss, ship in the corpus vectors.

# 6 Discussion

## 6.1 What the witnesses are — and are not

A witness signature certifies that *this wiring diagram, under this input, produced this raster*. Quorumed, it supports a claim that a sensed observation was processed through

a documented, replayable pipeline. It does not certify that an organism computed, relayed, or consented — the connectome is a preserved wiring *used as* computation; the organism is not claimed as the computer. This distinction is enforced in the protocol's lane table (living-system compute is refused) and in the claim labels.

### 6.2 The living cloud, honestly bounded

The architecture is designed so that a future E0+ deployment — sensors reading living systems, consent-gated emission, receipts crediting subjects — plugs into the receipt machinery demonstrated here without protocol changes. What this paper demonstrates is the verifiable spine: provenance that degrades into refusal rather than into trust, and a witness layer cheap enough to replay anywhere. Field validation is explicitly out of scope (Section 6.4).

### 6.3 Threat model

Appendix B details adversary classes; the load-bearing property is that *forgery requires either breaking BLAKE2s or defeating determinism*, while the honest-participant failure modes (mis-rasterized input, wrong blob version) are detectable at digest granularity. The residual trust point is the sensor itself — identical in kind to every remote-sensing system, and why the evidence gates exist upstream of the witnesses.

### 6.4 Limitations

(i) No field deployment: E0–E6 are pre-registered; none of the "living" lanes has produced a witnessed real-world receipt. (ii) Estimated joules: fresh benchmark runs use a power proxy; the measured-tier numbers in `docs/EFFICIENCY.md` come from RAPL-unlocked runs and are labelled accordingly. (iii) The witness layer validates *computation*, not *biology*: connectome verdicts carry no claim about cognition, ecology, or the health of the source organism. (iv) Sole-author project; the corpus is young. (v) Mesoscale and cellular graphs are mixed in one corpus deliberately — resolution is declared per row, and any use must respect it.

## 7 Availability

Code (AGPL-3.0): https://github.com/TerexitariusStomp/living-resources — `tools/connectome/`, `workers/compute-api/`, `tests/e2e/`. Corpus + vectors + spec: `paper/corpus/`, archived at https://doi.org/10.5281/zenodo.23681503 (release `paper-v1`). Reproduction commands: `paper/REPRODUCE.md`. Deployed API: https://templeearth.cc (`/verify` is a live replay endpoint). Claim ledger: `docs/SCIENCE.md`; pre-registered experiments: `docs/EXPERIMENTS.md`.

## Ethics and consent

No living system was instrumented, stimulated, or emitted at for this work; all connectome data are prior published reconstructions. Guardian-consent machinery described

here is implemented software, unexercised in the field. No animal experiments were performed.

## Funding and competing interests

Self-funded; no grant supported this work at submission. The author operates Temple Earth (Treetable LLC), the system described.

## Appendix A Formal lane and receipt specification

**Lanes.** $\mathcal{L} = \{\mathsf{sense}, \mathsf{fleet}, \mathsf{emit}, \mathsf{store}, \mathsf{hop}, \mathsf{compute}\}$. Gate map $g$: $\{\mathsf{sense}, \mathsf{fleet}\} \mapsto$evidence; $\{\mathsf{emit}, \mathsf{store}\} \mapsto$consent; $\{\mathsf{hop}, \mathsf{compute}\} \mapsto$refuse. Admission: $\mathsf{admit}(op) = \mathsf{admitted}(x)$ iff $op$.lane's gate verifies, else $\mathsf{refused}(r)$ — a signed value. **Consent tuple.** $C = \langle g, o, a, p \rangle$: guardian, operator, action class, policy hash; verified at admission, expiring, non-transferable. **Evidence claim.** $E = \langle s, \tau, \sigma \rangle$: source id, claim type, attestation; verifiable means checkable by a third party. **Witness verdict.** $W = \langle c, said_{in}, f, k, a, \gamma, s, d \rangle$: connectome, input SAID, feature vector, steps, action $\in \{-1, 0, +1\}$, confidence, spike count, digest. **Quorum.** $\geq 3$ distinct $c$ agreeing on $a$; output folds to $receipt = \mathsf{said}(\{W_i\}, policy, context)$ where $\mathsf{said}(x) =$"blake2s:" $\|$ blake2s(canon($x$)). **Encoder `lif-sensing-v1`** and blob layout are the normative text of `paper/corpus/SPEC.md` (reproduced there in full); constants: Q24 fixed-point, decay = 13,743,895, tonic 2,359,296, threshold 16,777,216, gain 25,165,824, weight-scale 7,190,235, variation-mod 1,677,720; rasterizer splits the population into affirm/refute halves driven by confidence, anomaly, agreement, and rate (equations in SPEC.md §4).

## Appendix B Threat model

**Adversary classes.** (a) *False detection*: a node claims an event that did not occur. Mitigation: the receipt binds the input SAID to a feature vector whose sources are themselves receipted; a forged claim needs forged upstream receipts — the same problem all MRV systems face, surfaced rather than hidden. (b) *Claimed replay*: asserting a digest without simulating. Defeated by public re-execution — the verification is the re-run. (c) *Implementation divergence*: a verifier drifts from the reference semantics. WiredReplay-1 vectors catch this at digest granularity. (d) *Quorum stuffing*: multiple verdicts from one wiring. Quorum counts *distinct connectome ids*. (e) *Consent bypass*: emitting without a tuple, or a tuple minted by the operator alone. The consent gate requires the guardian principal; emission is additionally duty-cycled by policy hash. (f) *Federated laundering*: importing an external receipt as local quorum. Federation marks provenance; refused by the lane table. **Residual trust**: the physical sensor and the guardian's key custody — universal to remote sensing, and the reason evidence gates sit upstream of witnesses.

## Appendix C Claim table

Condensed from `docs/SCIENCE.md` (authoritative):

- *Proven by science*: plant stress ultrasounds airborne + informative Khait et al (2023); biofilm $K^+$ waves Prindle et al (2015); biofilm coupling Liu et al (2017); electrical cell attraction Humphries et al (2017); reef acoustics track biodiversity Sueur and Farina (2015); C4D contactless conductivity Kubáň and Hauser (2009).
- *Built / live*: LIF-over-connectome replay with digest signatures (live: 15/15, parity 930/930); consent-gated emission stack (built, unexercised); quorum→receipt (live, 71/71 e2e); federation with provenance (built); mutual-credit settlement (built, zero credited detections to date).
- *Measured*: medulla crossover vs. dense fp32 (EFFICIENCY.md, RAPL-unlocked runs); fresh-run figures here are *estimated* tier. *Estimated*: all fresh-run joules. *Refused*: organism-as-computer, hop through living tissue, in-organism storage, ultrasound-as-instruction-set, unconsented emission, federated-to-quorum promotion.

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